

From operational soldering to quantum configuration space: Coherent deformations and projective comparison

Mingyang Han

School of Mathematical Sciences, Shanghai Jiao Tong University

Shanghai, 200240, China

hanmingyang@sjtu.edu.cn

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Abstract Starting from an operationally reconstructed Lorentzian solder map, we study the quantum kinematics associated with its local variations. At each event the ambient soldering space is $\mathcal{E}_x = \text{Hom}(T_x M, V_x)$, and the generator–displacement functions $\Theta_{\Pi, \xi}(e) = \Pi(e\xi)$ span its linear dual and separate configurations. We first show that an exact unitary implementation preserving additive composition independently in the generator and displacement arguments is necessarily Abelian, so the classical response alone does not generate noncommutative kinematics. We then impose one additional principle: soldering translations $e \mapsto e + A$ may be implemented coherently. The standard cotangent completion $T^*\mathcal{E}_x$ yields $[\widehat{K}(A), \widehat{\Theta}_{\Pi, \xi}] = i\hbar \Pi(A\xi)I$, while the response observables remain mutually commuting. On the nondegenerate sector, zero-level internal Lorentz reduction recovers the cotangent bundle of oriented, time-oriented Lorentz metric configuration space. As a secondary result, an anchored finite rotational comparison admits a rephasing-invariant diagnostic of spin-cover parity; the nontrivial parity is invisible to tensorial geometric observables and yields a conditional reducing-sector separation. The construction is local and kinematical, not a quantization of general relativity or a derivation of matter.

Keywords: operational soldering; quantum configuration space; soldering deformations; Weyl relations; Lorentz reduction; projective comparison

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1 Introduction

A basic choice in canonical quantization is the choice of configuration variables. In gravitational theories this choice is especially significant because the variables being quantized already encode the local geometry. Metrics, tetrads, soldering forms, connections, and holonomies have all served as classical or quantum variables in different formulations [21, 3, 4].

The question addressed here concerns the operational provenance of such a configuration variable. In the motion–staticity reconstruction program, the solder map is not introduced initially as a gravitational field. It arises from two preceding classical reconstruction steps.

The flat-space analysis of Ref. [15] starts from the bounded massive observer-state variables

$$(\mathbf{D}, S), \quad |\mathbf{D}|^2 + S^2 = 1, \quad S > 0,$$

and selects the standard relativistic massive-generator geometry through the corresponding operational and Hamiltonian rigidity results. The present paper uses the resulting internal Lorentz generator structure, rather than the bounded parametrization itself, as classical input.

The curved-space construction of Ref. [16] then begins with a smooth four-manifold M and an independent rank-four Lorentz vector bundle

$$V \longrightarrow M.$$

No identification

$$TM \simeq V$$

is assumed. Instead, scalar generator–displacement incidences are specified operationally. Under explicit composition, continuity, smoothness, calibration, and faithfulness assumptions, the response has a unique bilinear extension

$$\widetilde{\Theta}_x : V_x^* \times T_x M \longrightarrow \mathbb{R}$$

and reconstructs a unique nondegenerate solder map

$$e_x : T_x M \xrightarrow{\sim} V_x, \quad \widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi). \quad (1)$$

The Lorentzian metric is consequently

$$g = e^* \eta. \quad (2)$$

The present paper begins at this classical endpoint and asks what local quantum kinematics becomes available when the solder map itself is allowed to vary.

At a fixed event, all linear solderings form

$$\mathcal{E}_x := \text{Hom}(T_x M, V_x), \quad (3)$$

with nondegenerate sector

$$\mathcal{E}_x^\times := \text{Iso}(T_x M, V_x). \quad (4)$$

For fixed

$$\Pi \in V_x^*, \quad \xi \in T_x M,$$

the same scalar incidence expression defines a linear function on this configuration space,

$$\Theta_{\Pi, \xi} : \mathcal{E}_x \longrightarrow \mathbb{R}, \quad \Theta_{\Pi, \xi}(e) = \Pi(e\xi). \quad (5)$$

Finite-dimensional duality gives

$$\mathcal{E}_x^* \simeq V_x^* \otimes T_x M, \quad (6)$$

so finite linear combinations of the response functions exhaust the linear dual of soldering configuration space and separate its points. Thus the response that reconstructed one classical solder map becomes a complete linear configuration-observable family when e is allowed to vary.

The tensor-product identity Eq. (6) is standard. The structural point is its operational origin: the configuration observables are precisely the scalar generator–displacement incidences inherited from the preceding classical reconstruction. After cotangent completion, they constitute the configuration-observable sector associated with the standard vertical polarization of $T^* \mathcal{E}_x$.

Before conjugate variables are introduced, there is an important boundary. Suppose the classical response is represented unitarily while retaining exact additive composition independently in both the generator and displacement arguments. Section 2 shows that the resulting local response unitaries necessarily commute. The proof is an interchange argument of the same general type as the Eckmann–Hilton mechanism [11].

The corresponding exact-response algebra is therefore Abelian. Its spectral parameters may be represented by linear solder maps

$$e : T_x M \longrightarrow V_x,$$

but the algebra contains no operator that coherently changes one soldering configuration into another. Spectral uncertainty in e alone therefore does not produce a canonical noncommutative kinematics.

The additional quantum input introduced here is the *coherent soldering-deformation principle*: translations

$$e \longmapsto e + A, \quad A \in \mathcal{E}_x, \quad (7)$$

may be represented coherently. This principle is independent of both the classical soldering theorem and the exact-response algebra.

For the finite-dimensional linear configuration space $\mathcal{E}_x$, we then adopt the standard cotangent completion

$$\mathcal{P}_x = T^*\mathcal{E}_x. \quad (8)$$

Let

$$\mathbf{P} \in \mathcal{E}_x^*$$

denote the cotangent momentum conjugate to the solder map, thereby distinguishing it from the relativistic generator covector

$$\Pi \in V_x^*.$$

If $K(A)$ denotes the Hamiltonian generator of Eq. (7), the canonical pairing is

$$\{K(A), \Theta_{\Pi, \xi}\} = \Pi(A\xi). \quad (9)$$

Canonical quantization yields

$$[\widehat{K}(A), \widehat{\Theta}_{\Pi, \xi}] = i\hbar \Pi(A\xi)I. \quad (10)$$

The response observables remain mutually commuting. Noncommutativity enters through the operators that change the soldering configuration.

This distinguishes the local quantum construction from a direct “quantization of the response.” Exact implementation of the response alone remains Abelian; the noncommutative Weyl algebra arises only after the independent coherent-deformation principle is added.

The relation to the classical gravitational branch is correspondingly precise. In Ref. [16],

$$e \mapsto g = e^* \eta$$

and, under additional assumptions, the reconstructed geometry admits a restricted classical ADM/Einstein completion. The present paper does not quantize that completion. Instead, the same solder map is promoted to a variable in $\mathcal{E}_x$ and supplied with its cotangent complement:

$$\widetilde{\Theta} \longrightarrow e \begin{cases} \longrightarrow g \longrightarrow \text{restricted classical gravitational completion,} \\ \longrightarrow \mathcal{E}_x \longrightarrow T^*\mathcal{E}_x \longrightarrow \text{local quantum soldering kinematics.} \end{cases} \quad (11)$$

The second branch is the principal subject of this paper.

Internal Lorentz redundancy provides an important consistency check. On the nondegenerate sector,

$$\mathcal{E}_x^\times / SO^+(1, 3) \simeq \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M),$$

and zero-level cotangent reduction gives the corresponding cotangent metric configuration space. Thus the sixteen pre-reduction soldering coordinates reduce to the expected ten local metric coordinates rather than defining an unrelated geometric system.

A separate, secondary question concerns finite observer comparison. Once an operationally anchored rotational comparison is supplied, the observer stabilizer

$$SO(3)$$

may act projectively on a quantum comparison sector [5]. This projectivity is different from the local Weyl phase above. The Weyl phase is generated continuously by the symplectic pairing

$$\Pi(A\xi),$$

whereas rotational spin-cover parity is a global topological property associated with

$$\pi_1(SO(3)) \simeq \mathbb{Z}_2.$$

For the standard anchored comparison involving orthogonal π -rotations, we use the rephasing-invariant projective commutator

$$\Xi_\rho = U_\rho(r_1)U_\rho(r_2)U_\rho(r_1)^{-1}U_\rho(r_2)^{-1} \quad (12)$$

to diagnose the covering parity. In an irreducible lifted sector,

$$\Xi_\rho = \varsigma_\rho(-I)I, \quad \varsigma_\rho(-I) \in \{+1, -1\}.$$

The nontrivial sign is invisible to ordinary Lorentz vector and tensor representations. Under the additional compatibility assumptions stated in Sections 6 and 7, this yields a reducing decomposition of the tensorial geometric observable representation into the two spin-cover parities.

This latter result is intentionally narrower than a matter or superselection theorem. It does not produce a relativistic spinor field, select $j = \frac{1}{2}$, imply Fermi statistics, or determine a gauge group. Its role is to identify a global comparison datum that can remain invisible to the tensorial geometric algebra while being nontrivial in an enlarged quantum comparison theory.

The main logical chain of the paper is therefore

$$\begin{array}{c} \text{operational generator--displacement response} \\ \Theta_{\Pi, \xi} \end{array} \longrightarrow \mathcal{E}_x \longrightarrow T^*\mathcal{E}_x \longrightarrow \text{local Weyl quantum kinematics}, \quad (13)$$

with the cotangent step requiring the coherent soldering-deformation principle. The finite rotational comparison constitutes a separate conditional extension of this local construction.

The paper is organized as follows. Section 2 states the classical input and proves the exact-composition commutativity theorem. Section 3 shows that the operational response provides a complete linear configuration-observable family on soldering space. Section 4 introduces coherent soldering deformations and the cotangent/Weyl completion. Section 5 performs the internal Lorentz reduction and relates soldering variables to metric configuration space. Section 6 studies the independent global projectivity of anchored finite observer comparison, and Section 7 establishes the corresponding conditional spin-cover sector separation. Section 8 summarizes the scope and open problems.

2 Classical Input and the Exact-Response Boundary

This section isolates the classical structures used as input and then establishes the boundary encountered when the operational response is implemented exactly at the unitary level. The required classical results are those of Refs. [15, 16]; the commutativity and spectral statements below are established within the present paper.

2.1 Massive generator geometry and operational soldering

The flat-space input is the relativistic massive-generator geometry selected in Ref. [15]. Its detailed derivation from the bounded variables

$$(\mathbf{D}, S), \quad |\mathbf{D}|^2 + S^2 = 1, \quad S > 0,$$

is not required for the quantum construction below. We use only the resulting four-dimensional Lorentz vector spaces and their positive massive generator sectors.

Following the common convention of Refs. [15, 16], let

$$V \longrightarrow M$$

be a rank-four oriented and time-oriented real vector bundle equipped fiberwise with a Lorentzian metric

$$\eta_x$$

of signature

$$(-, +, +, +).$$

All vector–covector identifications, mass-shell formulas, and soldering relations in the present paper use this convention.

At this stage V remains conceptually distinct from TM .

For a fixed massive scale $Q > 0$, the positive-energy generator covector shell at x is

$$\Sigma_{Q,x}^+ = \left\{ \Pi \in V_x^* : \eta_x^*(\Pi, \Pi) = -Q^2, \quad \Pi \text{ on the positive-energy sheet} \right\}. \quad (14)$$

With the $(- + ++)$ convention and an oriented orthonormal internal frame, the associated four-momentum vector may be written

$$P^a = \left(\frac{E}{c}, \mathbf{p} \right),$$

while the generator covector used in the response is

$$\Pi_a = P_a^b = \left(-\frac{E}{c}, \mathbf{p} \right). \quad (15)$$

As in Ref. [16], we work in the common-observer sector identified in Ref. [15], in which the conversion scale c is shared by the massive species under consideration. No identification of this scale with the propagation speed of a particular quantum field is required here. Positive energy therefore means

$$-c \Pi_{\hat{0}} > 0.$$

Let

$$C_{Q,x} := \text{cone}(\Sigma_{Q,x}^+). \quad (16)$$

As shown in Ref. [16], this cone generates the full dual space,

$$C_{Q,x} - C_{Q,x} = V_x^*. \quad (17)$$

Generic elements of the cone away from the calibrated shell are composition variables; they are not additional fixed-mass one-particle states.

The primitive classical response is

$$\Theta_x : C_{Q,x} \times T_x M \longrightarrow \mathbb{R}. \quad (18)$$

Ref. [16] imposes additive composition and positive homogeneity in the generator argument, additive and continuous composition in the displacement argument, smooth base-point dependence, calibration, and operational faithfulness. Under these assumptions the response admits a unique bilinear extension

$$\widetilde{\Theta}_x : V_x^* \times T_x M \longrightarrow \mathbb{R}. \quad (19)$$

There is then a unique smooth bundle isomorphism

$$e : TM \xrightarrow{\sim} V \quad (20)$$

such that

$$\widetilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi) \quad (21)$$

for every

$$\Pi \in V_x^*, \quad \xi \in T_x M.$$

Its restriction to $C_{Q,x} \times T_x M$ is the original operational response.

The induced Lorentzian metric is

$$g_x(\xi, \zeta) = \eta_x(e_x \xi, e_x \zeta), \quad g = e^* \eta. \quad (22)$$

Operational faithfulness supplies the nondegeneracy required for e_x to be an isomorphism. The same reconstruction may be implemented tomographically using four linearly independent calibrated generator covectors. These are classical results of Ref. [16] and are taken as input here.

The distinction between Eqs. (18) and (19) will be maintained throughout. The symbol Θ_x denotes the original response on the positive generator cone, whereas $\widetilde{\Theta}_x$ denotes its bilinear extension to $V_x^* \times T_x M$. Whenever Π is allowed to range over the full vector space V_x^* , the extended response is understood.

The classical reconstruction selects one point

$$e_x \in \text{Iso}(T_x M, V_x)$$

from the operational incidence data. The question changes once the solder map itself is allowed to vary. The ambient linear space of possible solderings is

$$\mathcal{E}_x := \text{Hom}(T_x M, V_x), \quad (23)$$

with nondegenerate sector

$$\mathcal{E}_x^\times := \text{Iso}(T_x M, V_x). \quad (24)$$

For fixed $\Pi \in V_x^*$ and $\xi \in T_x M$, define

$$\Theta_{\Pi, \xi}(e) := \Pi(e \xi), \quad e \in \mathcal{E}_x. \quad (25)$$

Equation (25) is not yet a quantum observable assignment. It is the same incidence expression viewed as a linear function of a variable solder map. Its configuration-space meaning will be developed in Section 3.

2.2 Exact unitary composition and the Abelian boundary

We first ask whether the original composition laws can themselves generate a noncommutative quantum response algebra.

At a fixed event x , suppose that the response is represented by unitary operators

$$U_x(\Pi, \xi) \quad (26)$$

with

$$\Pi \in C_{Q,x}, \quad \xi \in T_x M.$$

Assume an exact implementation of generator composition,

$$U_x(\Pi_1 + \Pi_2, \xi) = U_x(\Pi_1, \xi)U_x(\Pi_2, \xi), \quad \Pi_1, \Pi_2 \in C_{Q,x}, \quad (27)$$

and an exact implementation of displacement composition,

$$U_x(\Pi, \xi + \zeta) = U_x(\Pi, \xi)U_x(\Pi, \zeta). \quad (28)$$

We also impose the natural normalization

$$U_x(0, \xi) = U_x(\Pi, 0) = I. \quad (29)$$

For unitary representations the corresponding identity normalization follows from the composition laws, but it is useful to state it explicitly.

The two exact composition requirements have an immediate consequence.

Theorem 2.1 (Exact-composition commutativity). *Let $U_x(\Pi, \xi)$ satisfy Eqs. (27)–(29). Then for every*

$$\Pi_1, \Pi_2 \in C_{Q,x}, \quad \xi, \zeta \in T_x M,$$

one has

$$[U_x(\Pi_1, \xi), U_x(\Pi_2, \zeta)] = 0. \quad (30)$$

Proof. Consider

$$U_x(\Pi_1 + \Pi_2, \xi + \zeta).$$

Applying generator composition first and displacement composition second gives

$$\begin{aligned} U_x(\Pi_1 + \Pi_2, \xi + \zeta) &= U_x(\Pi_1, \xi)U_x(\Pi_1, \zeta) \\ &\quad \times U_x(\Pi_2, \xi)U_x(\Pi_2, \zeta). \end{aligned} \quad (31)$$

Applying the two composition rules in the opposite order gives

$$\begin{aligned} U_x(\Pi_1 + \Pi_2, \xi + \zeta) &= U_x(\Pi_1, \xi)U_x(\Pi_2, \xi) \\ &\quad \times U_x(\Pi_1, \zeta)U_x(\Pi_2, \zeta). \end{aligned} \quad (32)$$

Equating Eqs. (31) and (32), multiplying from the left by $U_x(\Pi_1, \xi)^{-1}$, and multiplying from the right by $U_x(\Pi_2, \zeta)^{-1}$, one obtains

$$U_x(\Pi_1, \zeta)U_x(\Pi_2, \xi) = U_x(\Pi_2, \xi)U_x(\Pi_1, \zeta). \quad (33)$$

Since the generator and displacement arguments are arbitrary, the result follows. $\square$

The proof is an interchange argument: exact compatibility of the two independent additive structures enforces commutativity. This mechanism is closely related to the standard Eckmann–Hilton phenomenon [11]. Its role here is to identify the Abelian boundary of the specific generator–displacement response.

Let

$$\mathcal{A}_{\Theta,x} := W^* (\{U_x(\Pi, \xi)\}) \quad (34)$$

denote the von Neumann algebra generated by the exact response unitaries. Theorem 2.1 immediately gives the following result.

Corollary 2.2 (Abelian exact-response algebra). *The algebra $\mathcal{A}_{\Theta,x}$ is Abelian.*

Because the positive generator cone is cancellative and generates V_x^* , every

$$\Pi \in V_x^*$$

may be represented as

$$\Pi = \Pi_+ - \Pi_-, \quad \Pi_{\pm} \in C_{Q,x}.$$

For fixed ξ , define

$$U_x(\Pi, \xi) := U_x(\Pi_+, \xi)U_x(\Pi_-, \xi)^{-1}. \quad (35)$$

Exact generator composition together with Theorem 2.1 implies that this definition is independent of the chosen decomposition of Π . Thus, for each fixed ξ , the semigroup representation extends uniquely to the additive group V_x^* .

Assume from now on that the extended family

$$(\Pi, \xi) \mapsto U_x(\Pi, \xi)$$

is jointly strongly continuous. We use a joint spectral realization compatible with this continuity. Equivalently, the character relations may first be imposed on countable dense subsets of the finite-dimensional parameter spaces and then extended by continuity.

The resulting Abelian algebra has a useful spectral interpretation.

Proposition 2.3 (Spectral soldering parametrization). *Under the preceding exact-composition and strong-continuity assumptions, the Abelian response algebra admits a joint spectral representation such that, for almost every spectral point ω , there exists a linear map*

$$e_\omega \in \text{Hom}(T_x M, V_x) \quad (36)$$

for which

$$(U_x(\Pi, \xi)\psi)(\omega) = \exp\left[-\frac{i}{\hbar}\Pi(e_\omega\xi)\right]\psi(\omega). \quad (37)$$

Proof. By Corollary 2.2, the family of unitaries admits a joint spectral realization. At almost every spectral point ω , its scalar value

$$u_\omega(\Pi, \xi) \in U(1)$$

obeys the two exact composition laws.

For fixed ξ , $\Pi \mapsto u_\omega(\Pi, \xi)$ is a continuous character of the finite-dimensional real vector group V_x^* , while for fixed Π , $\xi \mapsto u_\omega(\Pi, \xi)$ is a continuous character of $T_x M$. Hence u_ω is a continuous bicharacter. Continuous bicharacters of finite-dimensional real vector groups are exponentials of real bilinear forms. Therefore there exists a real bilinear map

$$B_\omega : V_x^* \times T_x M \longrightarrow \mathbb{R}$$

such that

$$u_\omega(\Pi, \xi) = \exp\left[-\frac{i}{\hbar}B_\omega(\Pi, \xi)\right].$$

Finite-dimensional duality identifies every such bilinear form uniquely with a map

$$e_\omega : T_x M \longrightarrow V_x$$

through

$$B_\omega(\Pi, \xi) = \Pi(e_\omega\xi).$$

This gives Eq. (37). □

The proposition must be distinguished from the classical soldering theorem. In Ref. [16], operational faithfulness determines one nondegenerate classical map

$$e_x \in \mathcal{E}_x^\times.$$

By contrast, in Proposition 2.3, the maps e_ω are spectral parameters of an Abelian quantum algebra. They need not be nondegenerate unless the spectrum is explicitly restricted to $\mathcal{E}_x^\times$.

This distinction also shows why spectral uncertainty alone does not provide a quantum dynamics or a noncommutative quantum kinematics of geometry. A state of $\mathcal{A}_{\Theta,x}$ may have a nontrivial probability distribution over different spectral values e_ω , but every operator in the algebra remains diagonal in the common spectral representation. In particular, the exact response algebra contains no operator that coherently implements

$$e \mapsto e + A. \quad (38)$$

It therefore supplies no momentum conjugate to e and no Heisenberg pair on soldering configuration space.

The exact response algebra is consequently only a commuting configuration sector of a possible quantum theory. This conclusion follows directly from the simultaneous exact implementation of the two classical composition laws.

There are several inequivalent ways to enlarge this Abelian structure. One could relax exact composition to a projective composition law, introduce conjugate configuration-changing variables, or introduce path-dependent inter-event comparison operators. The classical response by itself does not uniquely select among these possibilities.

The present paper adopts one specific extension. It retains $\mathcal{A}_{\Theta,x}$ as a commuting configuration algebra and supplements it with coherent transformations between different values of e . Accordingly,

$$\mathcal{E}_x = \text{Hom}(T_x M, V_x)$$

will be treated in the next section as a configuration space rather than merely as a possible spectrum of the exact response algebra. The passage from a spectral label to a coherently transformable configuration variable is the additional quantum step on which the subsequent local construction depends.

3 Soldering Configuration Space and Response Coordinates

Section 2 established two logically distinct facts. First, the classical operational response of Ref. [16] determines one nondegenerate solder map

$$e_x : T_x M \xrightarrow{\sim} V_x.$$

Second, an exact unitary implementation of the same additive response laws remains Abelian and may be represented spectrally by linear solder maps, but it contains no operator that coherently moves between different soldering configurations.

We now isolate the configuration-space structure common to these two statements. The relevant finite-dimensional duality is standard. The specific point is that the response family inherited from the classical operational reconstruction supplies a complete linear family of configuration observables once the solder map itself is allowed to vary.

3.1 Response coordinates on soldering space

At a fixed event $x \in M$, define

$$\mathcal{E}_x := \text{Hom}(T_x M, V_x). \quad (39)$$

Since both vector spaces have dimension four,

$$\dim \mathcal{E}_x = 16. \quad (40)$$

The open subset

$$\mathcal{E}_x^\times := \text{Iso}(T_x M, V_x) \quad (41)$$

contains the nondegenerate solder maps selected by operational faithfulness. The larger vector space $\mathcal{E}_x$ will be used as the ambient configuration space because its linear structure allows soldering deformations to be defined without introducing an additional metric or connection on configuration space.

For every

$$\Pi \in V_x^*, \quad \xi \in T_x M,$$

define the linear function

$$\Theta_{\Pi, \xi} : \mathcal{E}_x \longrightarrow \mathbb{R}, \quad \Theta_{\Pi, \xi}(e) := \Pi(e\xi). \quad (42)$$

When $e = e_x$ is the classical solder map reconstructed in Ref. [16], Eq. (42) reproduces the extended operational response,

$$\Theta_{\Pi, \xi}(e_x) = \widetilde{\Theta}_x(\Pi, \xi). \quad (43)$$

The notation therefore distinguishes two roles: $\widetilde{\Theta}_x(\Pi, \xi)$ is the scalar response evaluated on the reconstructed classical configuration, whereas $\Theta_{\Pi, \xi}$ is the corresponding linear function on the full configuration space $\mathcal{E}_x$.

The relevant dual-space structure is immediate.

Proposition 3.1 (Dual of soldering configuration space). *There is a canonical vector-space isomorphism*

$$\mathcal{E}_x^* \simeq V_x^* \otimes T_x M. \quad (44)$$

Under this identification, the simple tensor $\Pi \otimes \xi$ acts on $e \in \mathcal{E}_x$ as

$$(\Pi \otimes \xi)(e) = \Pi(e\xi) = \Theta_{\Pi, \xi}(e). \quad (45)$$

Proof. For finite-dimensional vector spaces U and W ,

$$\text{Hom}(U, W) \simeq W \otimes U^*.$$

Taking the dual gives

$$\text{Hom}(U, W)^* \simeq W^* \otimes U.$$

Setting

$$U = T_x M, \quad W = V_x$$

gives Eq. (44). Under this identification, evaluation of the simple tensor $\Pi \otimes \xi$ on e is precisely Eq. (45). $\square$

The proposition immediately gives completeness of the linear response family.

Corollary 3.2 (Response-coordinate spanning). *The functions*

$$\{\Theta_{\Pi,\xi} : \Pi \in V_x^*, \xi \in T_x M\}$$

span the full linear dual:

$$\text{span}\{\Theta_{\Pi,\xi}\} = \mathcal{E}_x^*. \quad (46)$$

The significance of Corollary 3.2 is not the finite-dimensional tensor-product identity itself, but the origin of the functions appearing in Eq. (42). They are not arbitrary coordinates introduced after choosing $\mathcal{E}_x$; they are precisely the generator–displacement incidence functions inherited from the preceding operational reconstruction.

The same response family separates configurations.

Proposition 3.3 (Separation of soldering configurations). *Let*

$$e_1, e_2 \in \mathcal{E}_x.$$

If

$$\Theta_{\Pi,\xi}(e_1) = \Theta_{\Pi,\xi}(e_2) \quad (47)$$

for every

$$\Pi \in V_x^*, \quad \xi \in T_x M,$$

then

$$e_1 = e_2.$$

Proof. Equation (47) implies

$$\Pi((e_1 - e_2)\xi) = 0$$

for every

$$\Pi \in V_x^*, \quad \xi \in T_x M.$$

Since V_x^* separates points of V_x ,

$$(e_1 - e_2)\xi = 0$$

for every $\xi \in T_x M$. Hence

$$e_1 = e_2.$$

□

A finite coordinate realization follows immediately. Let

$$\{E_a\}_{a=0}^3$$

be a basis of V_x , with dual basis

$$\{E^a\}_{a=0}^3 \subset V_x^*,$$

and let

$$\{b_\mu\}_{\mu=0}^3$$

be a basis of $T_x M$. Write

$$e(b_\mu) = e^a{}_\mu E_a. \quad (48)$$

Then

$$\Theta_{E^a, b_\mu}(e) = e^a_{\mu}. \quad (49)$$

Thus sixteen members of the response family provide ordinary matrix coordinates on $\mathcal{E}_x$.

Equation (49) is basis dependent in form, but its content is intrinsic. Under simultaneous changes of internal and tangent bases, the sixteen coordinate functions transform as the components of an element of

$$V_x^* \otimes T_x M \simeq \mathcal{E}_x^*.$$

No preferred internal or tangent frame is selected.

It is useful to compare this statement with the finite tomography theorem of Ref. [16]. There, four independent calibrated massive generators determine one fixed classical solder map through the one-forms

$$\theta^{(A)}(\xi) = \Theta_x(\Pi^{(A)}, \xi).$$

Here the question is different. The solder map itself is allowed to vary, and the same incidence expressions become functions on the space of all linear solderings. Classical tomography reconstructs one point of $\mathcal{E}_x^\times$; the present construction identifies the complete linear observable content of the ambient configuration space.

These statements may be summarized as follows.

Theorem 3.4 (Operational response as a complete configuration family). *Let*

$$\mathcal{E}_x = \text{Hom}(T_x M, V_x),$$

and let

$$\Theta_{\Pi, \xi}(e) = \Pi(e\xi).$$

Then:

1. each $\Theta_{\Pi, \xi}$ is a linear function on $\mathcal{E}_x$;
2. finite linear combinations of these functions exhaust $\mathcal{E}_x^*$;
3. the complete response family separates points of $\mathcal{E}_x$;
4. in any chosen bases, sixteen response functions provide the ordinary matrix coordinates of a solder map.

Consequently, the generator–displacement response inherited from the classical reconstruction supplies a complete linear family of configuration observables on soldering space.

After the cotangent completion introduced in Section 4, these functions form the linear configuration-observable sector associated with the standard vertical polarization of $T^*\mathcal{E}_x$. Equivalently, their differentials span the pullback of $T_e^*\mathcal{E}_x$ to the cotangent phase space. The underlying finite-dimensional duality is standard; the construction-specific content is that this complete configuration family is supplied by the same operational response that reconstructed the classical solder map.

3.2 Configuration algebra and soldering deformations

Let

$$C_x^{\text{resp}} := \text{alg} \left\{ 1, \Theta_{\Pi, \xi} : \Pi \in V_x^*, \xi \in T_x M \right\} \quad (50)$$

denote the unital algebra generated under pointwise multiplication by the linear response functions on $\mathcal{E}_x$. Since these are ordinary scalar functions of the configuration e ,

$$[\Theta_{\Pi,\xi}, \Theta_{\Pi',\xi'}] = 0. \quad (51)$$

In a multiplication representation on a function space over $\mathcal{E}_x$, the corresponding operators likewise commute.

This is consistent with Theorem 2.1. The response functions therefore constitute a commuting configuration sector, not by themselves a complete quantum observable algebra.

The vector-space structure of $\mathcal{E}_x$ also provides canonical translation maps. For

$$A \in \mathcal{E}_x,$$

define

$$\tau_A : \mathcal{E}_x \longrightarrow \mathcal{E}_x, \quad \tau_A(e) = e + A. \quad (52)$$

At this stage Eq. (52) is only a transformation of the configuration space. No physical or quantum principle has yet been imposed requiring this transformation to be implemented coherently.

Its action on the response coordinates is fixed algebraically:

$$\begin{aligned} \Theta_{\Pi,\xi}(\tau_A(e)) &= \Pi((e + A)\xi) \\ &= \Theta_{\Pi,\xi}(e) + \Pi(A\xi). \end{aligned} \quad (53)$$

Therefore the constant vector field on $\mathcal{E}_x$ defined by A satisfies

$$\mathcal{L}_A \Theta_{\Pi,\xi} = \Pi(A\xi). \quad (54)$$

Equation (54) distinguishes a configuration readout from a configuration deformation. The labels (Π, ξ) specify a scalar readout of a solder map, whereas A specifies a tangent direction through soldering configuration space. Their natural evaluation pairing is

$$\langle \Theta_{\Pi,\xi}, A \rangle := \Pi(A\xi). \quad (55)$$

No metric on $\mathcal{E}_x$ is required.

The pairing is nondegenerate in the deformation argument.

Proposition 3.5 (Separation of deformation directions). *If*

$$\Pi(A\xi) = 0 \quad (56)$$

for every

$$\Pi \in V_x^*, \quad \xi \in T_x M,$$

then

$$A = 0.$$

Proof. For each fixed $\xi \in T_x M$, Eq. (56) gives

$$\Pi(A\xi) = 0$$

for every $\Pi \in V_x^*$. Since V_x^* separates points of V_x ,

$$A\xi = 0.$$

As this holds for every $\xi \in T_x M$, one has

$$A = 0.$$

□

Thus the response family detects every nonzero linear deformation of a soldering configuration. This statement is distinct from the operational faithfulness condition of Ref. [16]. Classical operational faithfulness states that the physical generator responses distinguish nonzero spacetime displacements for a fixed reconstructed solder map. Proposition 3.5 instead states that the full extended response-coordinate family separates nonzero tangent directions of the configuration space $\mathcal{E}_x$.

The construction is naturally fibered over spacetime. Define the bundle

$$\mathcal{E} := \text{Hom}(TM, V) \longrightarrow M, \quad (57)$$

whose fiber at x is $\mathcal{E}_x$. A classical solder field is a section

$$e \in \Gamma(\mathcal{E}), \quad (58)$$

and the nondegenerate classical sector lies in the open subbundle

$$\mathcal{E}^\times = \text{Iso}(TM, V). \quad (59)$$

The quantum construction developed here remains fiberwise. Passing from the finite-dimensional space $\mathcal{E}_x$ to a spacetime or spatial field configuration space such as

$$\Gamma(\mathcal{E})$$

requires structures not determined by the local response alone: an infinite-dimensional phase space, a class of admissible fields, regularization, constraints, boundary conditions, and a dynamical Hamiltonian or action. None of these is inferred from the finite-dimensional construction.

Likewise,

$$\dim \mathcal{E}_x = 16$$

is a configuration-space dimension and must not be interpreted as a count of propagating gravitational degrees of freedom. The nondegenerate sector contains an internal $SO^+(1, 3)$ redundancy, while no spacetime diffeomorphism reduction or gravitational constraint analysis has yet been imposed. The relation between soldering representatives and metric configurations will be treated separately in Section 5.

The logical content of this section may therefore be summarized as

$$\widetilde{\Theta}_x(\Pi, \xi) \longrightarrow \Theta_{\Pi, \xi} \in \mathcal{E}_x^* \longrightarrow \Pi(A\xi). \quad (60)$$

No Poisson bracket or quantum noncommutativity has yet been introduced. Section 4 adds one independent physical assumption: translations

$$e \longmapsto e + A$$

may be implemented coherently. Standard cotangent geometry then promotes the evaluation Eq. (55) to the canonical response–deformation bracket.

4 Cotangent Completion and Quantum Soldering Deformations

Section 3 identified the response family

$$\Theta_{\Pi, \xi}(e) = \Pi(e\xi)$$

as a complete linear family of configuration observables on

$$\mathcal{E}_x = \text{Hom}(T_x M, V_x).$$

No symplectic structure or quantum noncommutativity is required for that result. We now introduce the additional physical assumption that allows different soldering configurations to be connected coherently.

The cotangent-bundle construction used below is standard. Tetrad and soldering variables have long appeared as canonical variables in gravitational formulations [3, 4]. The role of the present construction is different: the configuration observables are supplied by the operational response reconstructed in the preceding classical work, and their pairing with soldering deformations is fixed by the scalar incidence

$$\Pi(A\xi).$$

4.1 Coherent deformations and the cotangent phase space

The additional assumption may be stated independently of a particular Hilbert-space representation.

Assumption 4.1 (Coherent soldering-deformation principle). *For every $A \in \mathcal{E}_x$, the configuration-space translation*

$$\tau_A(e) = e + A \tag{61}$$

may be represented as a coherent transformation between soldering configurations.

This assumption is independent of the classical operational soldering theorem. It is also not implied by the exact-response algebra of Section 2, whose elements act diagonally in a common spectral representation. It supplies precisely the configuration-changing structure absent at the exact-response boundary.

Assumption 4.1 does not uniquely select a quantum theory. We adopt the minimal standard cotangent completion of the linear configuration space,

$$\mathcal{P}_x := T^*\mathcal{E}_x \simeq \mathcal{E}_x \times \mathcal{E}_x^*. \tag{62}$$

This is a specified kinematical completion rather than a uniqueness statement about all possible quantum realizations of the operational response.

To distinguish the cotangent momentum from the relativistic generator covector

$$\Pi \in V_x^*,$$

we denote a momentum conjugate to the soldering configuration by

$$\mathbf{P} \in \mathcal{E}_x^*.$$

Choose a basis

$$\{E_a\}_{a=0}^3$$

of V_x , its dual basis

$$\{E^a\}_{a=0}^3,$$

and a basis

$$\{b_\mu\}_{\mu=0}^3$$

of $T_x M$, with dual basis $\{b^\mu\}$. Then

$$e = e^\mu_a E_a \otimes b^\mu, \quad \mathbf{P} = \mathbf{P}^\mu_a E^a \otimes b_\mu. \quad (63)$$

The canonical one-form on $T^* \mathcal{E}_x$ is

$$\vartheta_{\mathcal{E}} = \mathbf{P}^\mu_a de^\mu_a, \quad (64)$$

and we use

$$\Omega_{\mathcal{E}} = -d\vartheta_{\mathcal{E}} = de^\mu_a \wedge d\mathbf{P}^\mu_a. \quad (65)$$

Accordingly,

$$\{e^\mu_a, \mathbf{P}^\nu_b\} = \delta^\nu_a \delta^\mu_b. \quad (66)$$

For $A \in \mathcal{E}_x$, define the linear cotangent function

$$K(A) := -\mathbf{P}(A). \quad (67)$$

In components,

$$K(A) = -\mathbf{P}^\mu_a A^a_\mu. \quad (68)$$

We use the generator-first convention

$$\delta_A F := \{K(A), F\}. \quad (69)$$

Equation (66) then gives

$$\{K(A), e^\mu_a\} = A^a_\mu, \quad (70)$$

so that $K(A)$ generates the active translation

$$e \mapsto e + A.$$

The response coordinates are canonically paired with these deformation generators.

Proposition 4.2 (Canonical response–deformation pairing). *For every*

$$A \in \mathcal{E}_x, \quad \Pi \in V_x^*, \quad \xi \in T_x M,$$

one has

$$\{K(A), \Theta_{\Pi, \xi}\} = \Pi(A\xi). \quad (71)$$

Proof. In the bases above,

$$\Theta_{\Pi, \xi}(e) = \Pi_a \xi^\mu e^\mu_a,$$

while

$$K(A) = -\mathbf{P}^\mu_a A^a_\mu.$$

Using

$$\{\mathbf{P}^\mu_a, e^b_\nu\} = -\delta^b_a \delta^\mu_\nu,$$

we obtain

$$\begin{aligned} \{K(A), \Theta_{\Pi, \xi}\} &= -\Pi_a \xi^\mu A^b_\nu \{\mathbf{P}^\nu_b, e^\mu_a\} \\ &= \Pi_a A^a_\mu \xi^\mu \\ &= \Pi(A\xi). \end{aligned} \quad (72)$$

□

Equation (71) is the Hamiltonian realization of the directional derivative already obtained on configuration space,

$$\mathcal{L}_A \Theta_{\Pi, \xi} = \Pi(A\xi).$$

The symplectic completion therefore does not alter the operational incidence law. It promotes the previously identified response–deformation pairing to a canonical Poisson bracket.

More generally, for

$$F \in \mathcal{E}_x^*,$$

regarded as a linear configuration function,

$$\{K(A), F\} = F(A). \quad (73)$$

Since the response coordinates span $\mathcal{E}_x^*$, Eq. (71) contains the complete linear configuration–momentum pairing.

The nondegeneracy established in Proposition 3.5 now has a canonical interpretation:

$$A \neq 0 \implies \exists \Pi, \xi \text{ such that } \{K(A), \Theta_{\Pi, \xi}\} \neq 0. \quad (74)$$

Thus every nonzero constant soldering deformation changes at least one response coordinate.

The remaining linear brackets are

$$\{\Theta_{\Pi, \xi}, \Theta_{\Pi', \xi'}\} = 0, \quad (75)$$

$$\{K(A), K(B)\} = 0 \quad (76)$$

for constant $A, B \in \mathcal{E}_x$. The response functions therefore form a commuting configuration-observable sector, while the $K(A)$ form its conjugate translation sector.

There is an important operational qualification. The original physical response of Ref. [16] is directly defined only for generator covectors in the positive massive cone $C_{Q, x}$. The use of arbitrary $\Pi \in V_x^*$ in Eq. (71) refers to the unique linear extension of that response. Since the physical massive-generator sector spans V_x^* , this extension introduces no additional independent linear configuration directions.

4.2 Weyl quantization and the enlarged response algebra

The passage from Eq. (71) to canonical commutation relations is the standard quantization of a finite-dimensional linear phase space. It is useful to formulate the result first in Weyl form, where domain issues associated with unbounded operators are avoided.

For

$$A \in \mathcal{E}_x$$

and

$$F \in \mathcal{E}_x^*,$$

introduce unitary operators

$$U(A), \quad V(F),$$

satisfying

$$U(A)U(B) = U(A + B), \quad (77)$$

$$V(F)V(G) = V(F + G), \quad (78)$$

and the Weyl relation

$$U(A)V(F) = \exp\left(-\frac{i}{\hbar}F(A)\right)V(F)U(A). \quad (79)$$

This is the ordinary Weyl algebra associated with the canonical symplectic space

$$T^*\mathcal{E}_x.$$

The operationally inherited structure enters by taking

$$F = \Theta_{\Pi, \xi}.$$

For brevity, write

$$V(\Pi, \xi) := V(\Theta_{\Pi, \xi}).$$

Then

$$U(A)V(\Pi, \xi) = \exp\left(-\frac{i}{\hbar}\Pi(A\xi)\right)V(\Pi, \xi)U(A). \quad (80)$$

Thus the same scalar

$$\Pi(A\xi)$$

that describes the infinitesimal variation of an operational response in Section 3 appears as the central phase of the Weyl completion.

A regular representation yields the corresponding infinitesimal operators. Choose a translation-invariant Haar measure

$$de$$

on the finite-dimensional additive group $\mathcal{E}_x$, unique up to an overall positive constant, and define

$$\mathcal{H}_x = L^2(\mathcal{E}_x, de). \quad (81)$$

In the standard Schrödinger representation,

$$(U(A)\psi)(e) = \psi(e + A), \quad (82)$$

$$(V(F)\psi)(e) = \exp\left(-\frac{i}{\hbar}F(e)\right)\psi(e). \quad (83)$$

These equations give Eq. (79) directly.

On a suitable common dense invariant domain, the configuration and cotangent operators are

$$(\widehat{e}^a_\mu \psi)(e) = e^a_\mu \psi(e), \quad \widehat{P}^{a\mu}_a = -i\hbar \frac{\partial}{\partial e^a_\mu}, \quad (84)$$

and therefore

$$[\widehat{e}^a_\mu, \widehat{P}^{a\mu}_b] = i\hbar \delta^a_b \delta^\nu_\mu I. \quad (85)$$

Since

$$K(A) = -P(A),$$

the deformation generator acts as

$$(\widehat{K}(A)\psi)(e) = i\hbar A^a_\mu \frac{\partial \psi}{\partial e^a_\mu}(e), \quad (86)$$

while the operational response coordinate acts by multiplication,

$$(\widehat{\Theta}_{\Pi,\xi}\psi)(e) = \Pi_a \xi^\mu e^\mu \psi(e). \quad (87)$$

Consequently,

$$[\widehat{\Theta}_{\Pi,\xi}, \widehat{\Theta}_{\Pi',\xi'}] = 0, \quad (88)$$

$$[\widehat{K}(A), \widehat{K}(B)] = 0, \quad (89)$$

$$[\widehat{K}(A), \widehat{\Theta}_{\Pi,\xi}] = i\hbar \Pi(A\xi)I. \quad (90)$$

Equation (90) does not modify the original operational response law. The response observables remain mutually commuting. Noncommutativity enters only after the observable algebra is enlarged by operators that coherently change the soldering configuration.

The result may be collected as follows.

Theorem 4.3 (Operationally selected cotangent completion). *Let*

$$\mathcal{E}_x = \text{Hom}(T_x M, V_x)$$

carry the response functions

$$\Theta_{\Pi,\xi}(e) = \Pi(e\xi)$$

inherited from the classical operational soldering construction. If Assumption 4.1 is imposed and the standard cotangent completion

$$T^*\mathcal{E}_x$$

is adopted, then:

- 1.
2. *the response functions remain a commuting configuration-observable family associated with the standard vertical polarization of $T^*\mathcal{E}_x$;*
3. *every nonzero constant soldering deformation is detected by at least one response coordinate;*
4. *the response and deformation generators satisfy*

$$[\widehat{K}(A), \widehat{\Theta}_{\Pi,\xi}] = i\hbar \Pi(A\xi)I; \quad (91)$$

5. *the corresponding finite transformations satisfy*

$$U(A)V(\Pi, \xi) = \exp\left(-\frac{i}{\hbar} \Pi(A\xi)\right) V(\Pi, \xi)U(A); \quad (92)$$

6. *regular representations include the standard Schrödinger realization on $L^2(\mathcal{E}_x, de)$.*

Proof. The first statement follows from Eq. (88) and the configuration-space result of Theorem 3.4. The second follows from Proposition 3.5. Equation (91) is the canonical quantization of Eq. (71). The finite relation is Eq. (80), and Eqs. (81)–(87) provide the stated regular realization. $\square$

The theorem is conditional on the coherent-deformation principle and on the choice of the standard cotangent completion. Its framework-specific content is that the operational response selects the configuration-observable family and fixes the response–deformation pairing within that completion.

For states lying in a common domain of the infinitesimal generators, Eq. (90) gives the Robertson bound

$$\Delta K(A) \Delta \Theta_{\Pi, \xi} \geq \frac{\hbar}{2} |\Pi(A\xi)|. \quad (93)$$

A sharp response readout is therefore incompatible with a sharp conjugate deformation whenever

$$\Pi(A\xi) \neq 0.$$

This is the standard uncertainty relation for the corresponding canonical pair; its coefficient is fixed here by the operationally inherited generator–deformation incidence.

The enlarged algebra also makes precise how the exact-response boundary of Section 2 is crossed without contradicting it. Let

$$F = \Theta_{\Pi, \xi},$$

and let $E_F(\Delta)$ denote the spectral projection of $\widehat{F}$ associated with a Borel set $\Delta \subset \mathbb{R}$. Equation (82) gives

$$\mathbf{U}(A)\widehat{F}\mathbf{U}(A)^{-1} = \widehat{F} + F(A)I \quad (94)$$

and therefore

$$\mathbf{U}(A)E_F(\Delta)\mathbf{U}(A)^{-1} = E_F(\Delta - F(A)). \quad (95)$$

Whenever

$$F(A) = \Pi(A\xi) \neq 0,$$

the coherent deformation operator fails to commute with suitable spectral projections of the response observable.

Corollary 4.4 (Coherent departure from the exact-response algebra). *For every nonzero $A \in \mathcal{E}_x$, there exist*

$$\Pi \in V_x^*, \quad \xi \in T_x M$$

such that

$$[\mathbf{U}(A), E_{\Theta_{\Pi, \xi}}(\Delta)] \neq 0 \quad (96)$$

for suitable Borel sets Δ .

Proof. By Proposition 3.5, every nonzero A admits Π, ξ for which $\Pi(A\xi) \neq 0$. Equation (95) then shows that $\mathbf{U}(A)$ moves the spectral projections of the corresponding response observable. $\square$

The exact-response algebra is therefore retained as a commuting subalgebra; what changes is the larger algebra in which it is embedded. In the joint spectral representation of Section 2, the exact-response operators are diagonal in the soldering label. The operators $\mathbf{U}(A)$ act non-diagonally through

$$e \mapsto e + A$$

and provide coherent transformations between distinct soldering configurations.

Several qualifications are important.

First,

$$\mathcal{E}_x$$

contains degenerate as well as nondegenerate maps. Global translations on the ambient vector space do not preserve

$$\mathcal{E}_x^\times = \text{Iso}(T_x M, V_x).$$

For a fixed $e \in \mathcal{E}_x^\times$, sufficiently small translations remain in the open nondegenerate sector, whereas sufficiently large translations may cross the degeneracy locus. Quantization on all of $\mathcal{E}_x$ is therefore a linear kinematical completion; it does not identify every degenerate soldering with a physical gravitational configuration.

Second, the measure de in Eq. (81) is a Haar measure on the finite-dimensional additive configuration space. It is not a proposed physical measure for a quantum gravitational field theory.

Third, the construction remains fiberwise. Replacing

$$\mathcal{E}_x$$

by a space of sections such as

$$\Gamma(\mathcal{E})$$

produces an infinite-dimensional field-theoretic problem requiring regularization, constraints, boundary conditions, operator-domain choices, and dynamics not fixed by the local response.

Finally, because $\mathcal{E}_x$ is finite dimensional, the Stone–von Neumann theorem gives unitary equivalence of regular irreducible representations of the associated nondegenerate Heisenberg–Weyl relations with fixed nonzero central parameter [14]. This is only a kinematical uniqueness statement. It does not select a Hamiltonian, impose gravitational constraints, determine a vacuum, or extend automatically to the field-theoretic setting.

The local construction may therefore be summarized as

$$\Theta_{\Pi, \xi} \longrightarrow \mathcal{E}_x \longrightarrow T^*\mathcal{E}_x, \quad (97)$$

where the final step uses the coherent soldering-deformation principle and yields

$$[\widehat{K}(A), \widehat{\Theta}_{\Pi, \xi}] = i\hbar \Pi(A\xi)I. \quad (98)$$

The response family is inherited from the operational reconstruction and Section 3; the cotangent completion uses the additional Assumption 4.1. The next section restricts to the nondegenerate sector and separates internal Lorentz gauge directions from genuine changes of the reconstructed metric.

5 Lorentz Reduction and the Oriented Metric Configuration Space

Sections 3 and 4 were formulated on the ambient linear space

$$\mathcal{E}_x = \text{Hom}(T_x M, V_x).$$

This choice makes the response coordinates and coherent translations transparent, but not every direction in $\mathcal{E}_x$ represents a distinct physical geometry. On the nondegenerate sector, solder maps related by an internal Lorentz transformation induce the same spacetime metric.

The quotient below is standard frame and tetrad geometry, and its cotangent lift is a standard instance of symplectic reduction [4, 1, 20]. Its role in the present construction is to verify that the operationally selected configuration variables reduce to the expected local metric variables after the internal Lorentz redundancy is removed.

5.1 Nondegenerate solderings and the Lorentz metric quotient

Let

$$\mathcal{E}_x^\times = \text{Iso}(T_x M, V_x) \quad (99)$$

be the open nondegenerate subset of $\mathcal{E}_x$. The internal Lorentz space (V_x, η_x) carries the orientation and time orientation used in the classical reconstruction. Its proper orthochronous Lorentz group is

$$G_L := SO^+(V_x, \eta_x) \simeq SO^+(1, 3). \quad (100)$$

It acts on $\mathcal{E}_x^\times$ from the left,

$$e \mapsto \Lambda e, \quad \Lambda \in G_L. \quad (101)$$

The action is free. Indeed,

$$\Lambda e = e$$

with e invertible implies

$$\Lambda = I.$$

It is also proper. After choosing bases, $\mathcal{E}_x^\times$ is identified with $GL(4, \mathbb{R})$, and $SO^+(1, 3)$ is a closed subgroup acting by left multiplication. Hence the orbit space is a smooth manifold.

Every nondegenerate solder map induces a Lorentzian inner product on $T_x M$,

$$g_e := e^* \eta_x, \quad (102)$$

together with the orientation and time orientation transported from V_x .

Let

$$\text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M)$$

denote the space of Lorentzian inner products on $T_x M$ equipped with these two discrete structures. Define

$$q_x : \mathcal{E}_x^\times \longrightarrow \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M), \quad q_x(e) = (e^* \eta_x, \mathfrak{o}_e, \mathfrak{t}_e). \quad (103)$$

For $\Lambda \in G_L$,

$$(\Lambda e)^* \eta_x = e^* \eta_x, \quad (104)$$

and Λ preserves the internal orientation and time orientation. Hence q_x is constant on G_L -orbits.

Proposition 5.1 (Lorentz quotient of nondegenerate solderings). *The map q_x induces a natural diffeomorphism*

$$\mathcal{E}_x^\times / G_L \simeq \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M). \quad (105)$$

Proof. Let

$$(g, \mathfrak{o}, \mathfrak{t}) \in \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M).$$

Since $(T_x M, g)$ and (V_x, η_x) are four-dimensional Lorentz spaces with compatible orientation and time orientation, there exists an orientation- and time-orientation-preserving Lorentz isometry

$$e : (T_x M, g) \longrightarrow (V_x, \eta_x).$$

Thus q_x is surjective.

Suppose next that

$$q_x(e_1) = q_x(e_2).$$

Then

$$e_1^* \eta_x = e_2^* \eta_x$$

and the two solderings induce the same orientation and time orientation. The map

$$\Lambda := e_2 e_1^{-1} : V_x \longrightarrow V_x$$

preserves η_x , orientation, and time orientation. Hence

$$\Lambda \in G_L$$

and

$$e_2 = \Lambda e_1.$$

Therefore the fibers of q_x are precisely the G_L -orbits.

The smoothness of the induced map and its inverse follows from the usual frame-bundle description of Lorentzian inner products, equivalently from the free and proper group action above. $\square$

The dimension count is

$$\dim \mathcal{E}_x^\times - \dim SO^+(1, 3) = 16 - 6 = 10, \quad (106)$$

as expected for a symmetric bilinear form in four dimensions. This is a configuration-space dimension only. It is not a count of propagating gravitational degrees of freedom: no spacetime diffeomorphism reduction, Hamiltonian constraint, field equation, or dynamical gauge fixing has entered.

The same distinction is visible infinitesimally. Fix

$$e \in \mathcal{E}_x^\times$$

and let

$$\delta e \in T_e \mathcal{E}_x^\times.$$

Since e is invertible,

$$L := \delta e e^{-1} \in \text{End}(V_x). \quad (107)$$

The Lorentz form η_x gives the unique decomposition

$$L = L_- + L_+, \quad (108)$$

where

$$\eta_x(L_- u, v) = -\eta_x(u, L_- v), \quad (109)$$

$$\eta_x(L_+ u, v) = \eta_x(u, L_+ v). \quad (110)$$

Thus

$$L_- \in \mathfrak{so}(V_x, \eta_x) \simeq \mathfrak{so}(1, 3),$$

whereas L_+ is η_x -self-adjoint.

For

$$g = e^* \eta_x,$$

the induced metric variation is

$$\begin{aligned}\delta g(\xi, \zeta) &= \eta_x(\delta e \xi, e \zeta) + \eta_x(e \xi, \delta e \zeta) \\ &= 2\eta_x(L_+ e \xi, e \zeta).\end{aligned}\tag{111}$$

The Lorentz-antisymmetric part L_- therefore drops out of δg .

Hence

$$T_e \mathcal{E}_x^\times = \mathcal{V}_e \oplus \mathcal{S}_e,\tag{112}$$

where

$$\mathcal{V}_e = \{Xe : X \in \mathfrak{so}(V_x, \eta_x)\},\tag{113}$$

$$\mathcal{S}_e = \{Se : S^{\dagger\eta} = S\}.\tag{114}$$

The vertical space $\mathcal{V}_e$ is tangent to the internal Lorentz gauge orbit, whereas $\mathcal{S}_e$ maps isomorphically to

$$T_{g_e} \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M) \simeq \text{Sym}^2(T_x^* M).$$

This splitting has a direct consequence for the response coordinates. A general deformation

$$A \in T_e \mathcal{E}_x^\times$$

may be written

$$A = Xe + Se,$$

with

$$X \in \mathfrak{so}(V_x, \eta_x), \quad S^{\dagger\eta} = S.$$

The canonical response–deformation pairing becomes

$$\Pi(A\xi) = \Pi(Xe\xi) + \Pi(Se\xi).\tag{115}$$

Before Lorentz reduction the response coordinates resolve both changes of internal representative and changes of the induced metric. After reduction, only the symmetric component represents a tangent direction of metric configuration space.

For later comparison with the rotational sector, choose a future-directed unit timelike vector

$$y \in T_x M, \quad g(y, y) = -1,$$

and define

$$H_y := y^\perp.\tag{116}$$

The stabilizer of y is

$$SO(H_y) \simeq SO(3).$$

Relative to this observer, a symmetric metric variation decomposes under $SO(3)$ as

$$\text{Sym}^2(T_x^* M) \simeq \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{1} \oplus \mathbf{5}.\tag{117}$$

This is the standard observer-frame decomposition of the ten metric directions. It neither turns the observer rotation group into an independent physical symmetry nor introduces projective quantum structure.

5.2 Cotangent reduction and the status of the quantum representation

The action

$$e \mapsto \Lambda e$$

on $\mathcal{E}_x^\times$ has the canonical cotangent lift to

$$T^*\mathcal{E}_x^\times.$$

With the notation introduced in Section 4, a phase-space point is

$$(e, P), \quad P \in \mathcal{E}_x^*.$$

The lifted action is

$$(e, P) \mapsto (\Lambda e, \Lambda \cdot P), \quad (118)$$

where the induced dual action is defined by

$$(\Lambda \cdot P)(A) := P(\Lambda^{-1}A), \quad A \in \mathcal{E}_x. \quad (119)$$

The cotangent lift preserves the canonical one-form and symplectic form.

For

$$X \in \mathfrak{so}(V_x, \eta_x) \simeq \mathfrak{so}(1, 3),$$

the infinitesimal configuration-space action is

$$X_{\mathcal{E}}(e) = Xe.$$

The standard cotangent-lift momentum map

$$J : T^*\mathcal{E}_x^\times \longrightarrow \mathfrak{so}(1, 3)^* \quad (120)$$

is characterized by

$$\langle J(e, P), X \rangle = P(Xe). \quad (121)$$

An overall sign changes if the opposite Hamiltonian-generator convention is adopted; the zero level used below is unaffected.

The condition

$$J(e, P) = 0 \quad (122)$$

is equivalent to

$$P(Xe) = 0 \quad \text{for every } X \in \mathfrak{so}(1, 3). \quad (123)$$

Thus P annihilates the vertical Lorentz directions and descends to a covector on the metric quotient.

The standard zero-level cotangent reduction theorem now applies [1, 20].

Proposition 5.2 (Zero-level Lorentz reduction). *For the free and proper G_L -action on $\mathcal{E}_x^\times$, there is a natural symplectomorphism*

$$J^{-1}(0)/G_L \simeq T^*(\mathcal{E}_x^\times/G_L) \simeq T^*\text{Met}_{1,3}^{\text{or},\uparrow}(T_x M). \quad (124)$$

Proof. For a free and proper action on a configuration manifold Q , the zero level of the cotangent-lift momentum map is the annihilator of the vertical bundle. Quotienting this annihilator by the group gives canonically

$$T^*(Q/G).$$

Taking

$$Q = \mathcal{E}_x^\times$$

and using Eq. (123) gives the first identification. Proposition 5.1 gives the second. $\square$

Equation (124) is standard symplectic reduction. In the present construction it shows that the cotangent variables introduced by the coherent soldering completion reduce, at zero internal Lorentz momentum, to the cotangent variables of the local metric configuration space.

This relation can be written explicitly. Choose bases and write

$$g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu. \quad (125)$$

Let e_a^μ denote the inverse soldering matrix,

$$e_a^\mu e^a_\nu = \delta^\mu_\nu, \quad e_a^\mu e^b_\mu = \delta^b_a,$$

and define

$$e_{a\mu} := \eta_{ab} e^b_\mu, \quad e^{a\mu} := \eta^{ab} e_b^\mu.$$

Let

$$p^{\mu\nu} = p^{\nu\mu}$$

be a covector conjugate to the symmetric metric components. Then

$$\delta g_{\mu\nu} = e_{a\nu} \delta e^a_\mu + e_{a\mu} \delta e^a_\nu, \quad (126)$$

so that

$$p^{\mu\nu} \delta g_{\mu\nu} = 2 p^{\mu\nu} e_{a\nu} \delta e^a_\mu. \quad (127)$$

The horizontal soldering momentum corresponding to $p^{\mu\nu}$ is therefore

$$P_a^\mu = 2 p^{\mu\nu} e_{a\nu}. \quad (128)$$

Conversely, for a general P_a^μ , define its symmetric metric projection by

$$p^{\mu\nu} = \frac{1}{4} (P_a^\mu e^{a\nu} + P_a^\nu e^{a\mu}). \quad (129)$$

On $J^{-1}(0)$, the quantity

$$P_a^\mu e^{a\nu}$$

is symmetric in μ, ν , and hence

$$p^{\mu\nu} = \frac{1}{2} P_a^\mu e^{a\nu}. \quad (130)$$

The antisymmetric part of the unreduced soldering momentum is precisely the part paired with internal Lorentz gauge directions.

This reduction also clarifies the status of the response observables. For fixed labels $\Pi \in V_x^*$ and $\xi \in T_x M$, the function

$$\Theta_{\Pi, \xi}(e) = \Pi(e\xi)$$

is Lorentz covariant rather than individually gauge invariant. Under a change of internal representative

$$e \mapsto \Lambda e,$$

one has

$$\Theta_{\Pi, \xi}(\Lambda e) = \Theta_{\Lambda^* \Pi, \xi}(e), \quad (\Lambda^* \Pi)(v) := \Pi(\Lambda v). \quad (131)$$

Equivalently, if the generator covector is transformed simultaneously as

$$\Pi \mapsto (\Lambda^{-1})^* \Pi, \quad (132)$$

then

$$\Theta_{(\Lambda^{-1})^* \Pi, \xi}(\Lambda e) = \Theta_{\Pi, \xi}(e). \quad (133)$$

The scalar incidence is therefore independent of the internal Lorentz representative when both internal objects are transformed consistently, although a component carrying a fixed internal label remains frame dependent.

The pre-reduction Schrödinger representation

$$L^2(\mathcal{E}_x, de) \quad (134)$$

from Section 4 must accordingly be interpreted as a representation of the ambient finite-dimensional Weyl algebra, not as a physical Hilbert space for gravity. Classical Lorentz reduction is described by Eq. (124); quantization after reduction is a separate problem.

Formally, infinitesimal Lorentz invariance could be represented by operators

$$\widehat{J}_X, \quad X \in \mathfrak{so}(1, 3),$$

with a constraint of the schematic form

$$\widehat{J}_X \Psi = 0. \quad (135)$$

We do not identify the solution space of Eq. (135) with a physical Hilbert space. The gauge group $SO^+(1, 3)$ is noncompact, and a reduced quantum Hilbert space requires choices of measure, operator domains, treatment of gauge orbits, and possibly refined or group-averaged quantization. Even after internal Lorentz reduction, spacetime diffeomorphism and dynamical gravitational constraints remain outside the present fiberwise construction.

A second limitation is already visible classically. The ambient translation

$$e \mapsto e + A$$

used in the coherent deformation principle does not preserve $\mathcal{E}_x^\times$ globally. The Weyl representation on all of $\mathcal{E}_x$ and the Lorentz-reduced phase space on $\mathcal{E}_x^\times$ therefore answer different kinematical questions. Locally around a nondegenerate soldering, sufficiently small deformations remain in $\mathcal{E}_x^\times$; globally, a theory restricted strictly to nondegenerate geometry requires additional control of the degeneracy locus.

The logical role of this section may be summarized as

$$\begin{array}{ccccc} \text{response-selected cotangent kinematics} & \longrightarrow & \text{internal Lorentz reduction} & \longrightarrow & T^* \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M). \\ T^* \mathcal{E}_x & & J = 0 & & \end{array} \quad (136)$$

The arrows use standard frame and symplectic geometry. Their significance here is that the configuration variables selected by the operational response reduce to the expected metric configuration variables rather than forming an unrelated sixteen-component system.

The internal Lorentz quotient concerns gauge redundancy of the local geometric representative. It should be distinguished from the projective realization of a relationally specified finite observer comparison. The next section addresses that separate, secondary question.

6 Global Projective Comparison and a Spin-Cover Diagnostic

The canonical construction of Section 4 produces a local continuous source of quantum projectivity through the Weyl relation

$$U(A)V(\Pi, \xi) = \exp\left(-\frac{i}{\hbar}\Pi(A\xi)\right)V(\Pi, \xi)U(A).$$

Its phase is determined by the symplectic pairing on $T^* \mathcal{E}_x$.

A distinct source of projectivity may arise in finite observer comparisons. Because quantum states are represented by rays, a classically ordinary comparison transformation may act projectively on a quantum comparison sector. The relevant representation theory, including the double covering $SU(2) \rightarrow SO(3)$, is standard [5]. The purpose of this section is to distinguish this global projective structure from the local Weyl projectivity above and to identify a finite comparison diagnostic for its covering parity.

The operational status of the spinorial phase under finite rotations has a substantial history. Aharonov and Susskind proposed that the sign change of a spinor under a 2π rotation can have observable consequences; Hegerfeldt and Kraus subsequently analyzed limitations of that argument, while neutron interferometry later directly observed the corresponding fermionic phase reversal [2, 17, 22].

Related modern work on quantum reference frames has developed operational and relational descriptions of reference-frame transformations, relativistic spin, and general symmetry-group changes of perspective [6, 12, 7, 19]. The anchored comparison assumption used below is more specific: it provides the relational comparison structure required for the finite projective diagnostic, without identifying local Lorentz gauge rotations themselves as observables.

6.1 Anchored rotational comparison and its projective classes

Let

$$g$$

be a Lorentz metric represented by a point of the reduced configuration space of Section 5, and choose a future-directed unit timelike observer vector

$$y \in T_x M, \quad g(y, y) = -1. \quad (137)$$

Its orthogonal spatial subspace is

$$H_y := y^\perp, \quad (138)$$

with Euclidean metric induced by g . The proper spatial rotation group is

$$G_y := SO(H_y) \simeq SO(3). \quad (139)$$

An arbitrary local Lorentz-frame rotation is an internal gauge transformation of the soldering representative and is not, by itself, an observable comparison. A projective comparison becomes physically meaningful only after a protocol has been specified that relates distinct frame orientations to operationally distinguishable comparison histories.

We therefore state this additional input explicitly.

Assumption 6.1 (Anchored rotational comparison). *A relational, asymptotic, interferometric, or otherwise operationally anchored comparison prescription is available such that the observer rotation group*

$$G_y \simeq SO(3)$$

acts effectively on a quantum comparison sector $\mathcal{H}_\rho$. The induced action on rays is strongly continuous.

This assumption is independent of the local soldering reconstruction, Lorentz gauge reduction, and the coherent soldering-deformation principle. The discussion below determines the projective information carried by such a comparison sector once it exists.

A strongly continuous action on rays may be represented by a projective unitary representation

$$\overline{U}_\rho : G_y \longrightarrow PU(\mathcal{H}_\rho). \quad (140)$$

After choosing unitary representatives

$$U_\rho(r) \in U(\mathcal{H}_\rho),$$

one has

$$U_\rho(r)U_\rho(s) = c_\rho(r, s)U_\rho(rs), \quad c_\rho(r, s) \in U(1). \quad (141)$$

Associativity gives

$$c_\rho(r, s)c_\rho(rs, t) = c_\rho(s, t)c_\rho(r, st), \quad (142)$$

while the rephasing

$$U_\rho(r) \longmapsto e^{i\alpha(r)}U_\rho(r) \quad (143)$$

changes the multiplier by a coboundary without changing the projective representation.

For $SO(3)$, the relevant global structure is the universal covering sequence

$$1 \longrightarrow \{I, -I\} \longrightarrow SU(2) \xrightarrow{p} SO(3) \longrightarrow 1. \quad (144)$$

Equivalently,

$$\pi_1(SO(3)) \simeq \mathbb{Z}_2. \quad (145)$$

Under the usual regularity assumptions, the projective representation lifts to an ordinary unitary representation

$$\widetilde{U}_\rho$$

of $SU(2)$. In an irreducible sector the central covering element

$$z := -I$$

acts by a scalar,

$$\widetilde{U}_\rho(z) = \varsigma_\rho(z)I, \quad \varsigma_\rho(z) \in \{+1, -1\}. \quad (146)$$

The value

$$\varsigma_\rho(z) = +1$$

corresponds to a representation that descends to an ordinary unitary representation of $SO(3)$, whereas

$$\varsigma_\rho(z) = -1$$

corresponds to the nontrivial spin-cover projective class.

This global projectivity is mathematically distinct from the Weyl phase of Section 4. The latter arises from the symplectic pairing between configuration and cotangent directions. The rotation-group obstruction instead originates from the global topology of $SO(3)$. Thus

$$\begin{aligned} \text{local canonical projectivity} & : \exp\left(-\frac{i}{\hbar}\Pi(A\xi)\right), \\ \text{global rotational projectivity} & : \varsigma_\rho(-I) \in \{+1, -1\}. \end{aligned} \quad (147)$$

Neither structure is a reformulation of the other.

In particular, the canonical commutator

$$[\widehat{K}(A), \widehat{\Theta}_{\Pi, \xi}] = i\hbar \Pi(A\xi)I$$

already provides noncommutative quantum kinematics even if every finite rotational comparison sector is tensorial. Therefore

$$\text{canonical noncommutativity} \not\Rightarrow \text{spin-cover projectivity}. \quad (148)$$

A nontrivial rotational projective sector is an additional representation-theoretic or dynamical possibility.

6.2 Orthogonal π -rotation comparison and the diagnostic Ξ

We now specify a finite comparison protocol that detects the rotational projective class without assigning a spin label as part of the input.

Choose an oriented orthonormal basis

$$(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$$

of H_y , and let

$$r_1, \quad r_2, \quad r_3$$

be rotations through angle π about the corresponding axes. In the defining three-dimensional representation,

$$r_1 = \text{diag}(1, -1, -1), \quad (149)$$

$$r_2 = \text{diag}(-1, 1, -1), \quad (150)$$

$$r_3 = \text{diag}(-1, -1, 1). \quad (151)$$

They satisfy

$$r_1 r_2 = r_2 r_1 = r_3, \quad (152)$$

with cyclic analogues. Thus

$$\{I, r_1, r_2, r_3\}$$

is the Klein four subgroup

$$V_4 \simeq \mathbb{Z}_2 \times \mathbb{Z}_2 \subset SO(3).$$

Choose projective unitary representatives

$$U_i := U_\rho(r_i).$$

Since r_1 and r_2 commute classically, define their projective commutator

$$\Xi_\rho := U_1 U_2 U_1^{-1} U_2^{-1}. \quad (153)$$

For any commuting pair in a projective representation, this commutator is rephasing invariant. In the present case,

$$\Xi_\rho = \frac{c_\rho(r_1, r_2)}{c_\rho(r_2, r_1)} I. \quad (154)$$

Its role here is as a finite operational diagnostic associated with the specified observer-comparison protocol.

Indeed,

$$U_i \mapsto e^{i\alpha_i} U_i \quad \implies \quad \Xi_\rho \mapsto \Xi_\rho, \quad (155)$$

so Ξ_ρ is independent of the arbitrary phases chosen for the projective representatives.

Its value is most transparent in the universal cover. Let

$$\alpha_i : [0, 1] \longrightarrow SO(3), \quad \alpha_i(t) = R_{e_i}(\pi t),$$

be the standard continuous paths from the identity to r_i . Choose their lifts beginning at $I \in SU(2)$. Their endpoints may be represented in the defining spinor representation as

$$\widetilde{r}_i = -i\sigma_i, \quad (156)$$

where σ_i are the Pauli matrices. Hence

$$\widetilde{r}_1 \widetilde{r}_2 = -i\sigma_3, \quad (157)$$

$$\widetilde{r}_2 \widetilde{r}_1 = +i\sigma_3 = -\widetilde{r}_1 \widetilde{r}_2. \quad (158)$$

Therefore

$$\widetilde{r}_1 \widetilde{r}_2 \widetilde{r}_1^{-1} \widetilde{r}_2^{-1} = -I. \quad (159)$$

The same result has a path interpretation. Define one path from I to $r_1 r_2$ by following α_1 and then the left-translated path

$$r_1 \alpha_2,$$

and another by following α_2 and then

$$r_2 \alpha_1.$$

Since

$$r_1 r_2 = r_2 r_1,$$

the two paths have the same endpoints. Concatenating the first with the reverse of the second produces a closed loop

$$\lambda_{\text{rot}} : S^1 \longrightarrow SO(3). \quad (160)$$

For the standard paths above, the lift of λ_{rot} beginning at I terminates at $-I$. The loop therefore represents the nontrivial element of $\pi_1(SO(3))$.

The explicit comparison paths matter. Equality of the endpoints

$$r_1 r_2 = r_2 r_1$$

alone does not determine the homotopy class of arbitrary comparison histories. The diagnostic below therefore refers to the standard orthogonal- π protocol just defined, or to an anchored protocol homotopic to it relative to the endpoints.

For an irreducible spin- j representation,

$$\rho_j(-I) = (-1)^{2j} I.$$

Combining this relation with Eq. (159) gives

$$\Xi_j = (-1)^{2j} I. \quad (161)$$

Hence

$$\Xi_j = +I, \quad j \in \mathbb{Z}, \quad (162)$$

$$\Xi_j = -I, \quad j \in \mathbb{Z} + \frac{1}{2}. \quad (163)$$

It is more useful for the present construction to formulate the result in terms of the projective class rather than a spin label.

Proposition 6.2 (Orthogonal- π comparison diagnostic). *Assume Assumption 6.1 and use the standard orthogonal- π comparison paths Eq. (6.2). Then the rephasing-invariant operator Ξ_ρ defined in Eq. (153) represents the covering holonomy of the associated nontrivial loop in $SO(3)$. In an irreducible comparison sector,*

$$\Xi_\rho = \widetilde{U}_\rho(-I) = \varsigma_\rho(-I)I. \quad (164)$$

Consequently, $\Xi_\rho = -I$ detects the nontrivial spin-cover projective class, whereas $\Xi_\rho = +I$ corresponds to the tensorial class.

Proof. The projective commutator is invariant under rephasing by Eq. (155). The standard comparison loop lifts from I to the nontrivial covering element $-I$ by Eq. (159). The projective commutator therefore represents the covering holonomy. In an irreducible lifted representation the central element acts as $\varsigma_\rho(-I)I$, which gives the result. $\square$

The finite subgroup underlying the calculation may also be displayed explicitly. The inverse image of

$$V_4 \subset SO(3)$$

under the covering map is the quaternion group

$$Q_8 = \{\pm I, \pm i\sigma_1, \pm i\sigma_2, \pm i\sigma_3\}, \quad (165)$$

with central extension

$$1 \longrightarrow \mathbb{Z}_2 \longrightarrow Q_8 \longrightarrow V_4 \longrightarrow 1. \quad (166)$$

This makes the diagnostic transparent: the classical comparison subgroup is Abelian, while its lift can retain the nontrivial central sign.

The value

$$\Xi_\rho = -I$$

determines only the parity of the rotational projective class. It does not select a unique spin. All

$$j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$$

give the same covering sign. If irreducibility and minimal carrier dimension are imposed separately, the smallest nontrivial representation is $j = \frac{1}{2}$; this is an additional representation-theoretic criterion.

Likewise, the rotational result does not by itself construct a covariant relativistic spinor field. The universal cover

$$Spin^+(1, 3) \simeq SL(2, \mathbb{C})$$

has the standard finite-dimensional modules

$$(j_L, j_R),$$

with covering center acting as

$$(-1)^{2(j_L + j_R)} I.$$

The Weyl modules

$$\left(\frac{1}{2}, 0\right), \quad \left(0, \frac{1}{2}\right)$$

are the smallest covariant modules with nontrivial center action, but they are not finite-dimensional unitary representations of the noncompact Lorentz group. Passing from the rotational comparison sector to a relativistic spinor field requires additional field-theoretic structure.

Nor does

$$\Xi_\rho = -I$$

imply Fermi exchange statistics. The spin–statistics connection requires relativistic locality, positivity or spectrum assumptions, and the hypotheses of an appropriate spin–statistics theorem [13]. The diagnostic here identifies only the rotational spin-cover class.

Finally, the nontrivial class is permitted rather than forced. Neither the operational soldering theorem nor the coherent deformation principle requires

$$\varsigma_\rho(-I) = -1.$$

The local quantum soldering algebra remains well defined when all finite comparison sectors are tensorial.

The logical structure of this section is

$$\begin{array}{ccccc} \text{anchored observer comparison} & \longrightarrow & \text{projective comparison sector} & \longrightarrow & \text{orthogonal-}\pi \text{ diagnostic} \\ G_y \simeq SO(3) & & \overline{U}_\rho & & \Xi_\rho = \varsigma_\rho(-I)I. \end{array} \quad (167)$$

The physical input is the anchored comparison assumption. The covering classification then supplies a finite diagnostic that is independent of the local canonical projectivity of soldering deformations.

The covering element detected by Ξ_ρ projects to the identity in the ordinary Lorentz group and is therefore invisible to vector and tensor representations of the reconstructed geometry. The next section states the corresponding relative sector-separation result.

7 Tensorial Geometric Algebra and Spin-Cover Sector Separation

The covering-parity separation considered below is closely related to the classical univalence superselection problem [23, 18, ?]. Here the issue is specialized to the tensorial geometric algebra selected by the operational soldering construction and to the covering character identified by the anchored comparison diagnostic.

Section 6 identified the conditional comparison datum

$$\Xi_\rho = \varsigma_\rho(-I)I$$

for an anchored rotational comparison sector. We now ask what the nontrivial value

$$\Xi_\rho = -I$$

implies relative to an algebra generated by the reconstructed tensorial geometry.

The key representation-theoretic fact is that the nontrivial element of the spin-cover kernel projects to the identity in the ordinary Lorentz group and therefore acts trivially on vector and tensor representations. If the same central element acts nontrivially in an enlarged comparison sector, the tensorial geometric observable algebra cannot connect the two covering parities. The resulting statement is a relative kinematical sector separation, not a full superselection theory of quantum fields.

7.1 Spin-cover center and the tensorial geometric algebra

Let

$$p : Spin^+(1, 3) \longrightarrow SO^+(1, 3) \quad (168)$$

be the double covering, with kernel

$$\ker p = \{1, z\}, \quad z = -1. \quad (169)$$

Thus

$$z^2 = 1, \quad p(z) = I.$$

We distinguish a tensorial geometric field algebra from its gauge-invariant observable subalgebra. Let

$$\mathfrak{F}_{\text{geom}}^{\text{tens}} \quad (170)$$

denote a represented algebra generated by geometric quantities whose Lorentz transformation laws factor through ordinary tensor representations of $SO^+(1, 3)$. Depending on the formulation, this may include the soldering field, tensorial curvature and torsion components, the metric, and tensorial functions constructed from them. Gauge-covariant elements of $\mathfrak{F}_{\text{geom}}^{\text{tens}}$ need not individually be observables.

Let

$$\mathcal{A}_{\text{geom}} \subseteq \mathfrak{F}_{\text{geom}}^{\text{tens}} \quad (171)$$

denote the represented gauge-invariant geometric observable algebra relevant to the present kinematical discussion. No local observable net is constructed here. The only property needed below is that the spin-group action on these geometric variables factors through the covering map p .

Let

$$Spin(3)_y \subset Spin^+(1, 3)$$

denote the spin lift of the observer stabilizer

$$SO(H_y) \simeq SO(3).$$

Thus

$$Spin(3)_y \simeq SU(2),$$

and its nontrivial covering-kernel element is the same central element

$$z = -1$$

of $Spin^+(1, 3)$.

Assume that the enlarged quantum comparison theory admits a unitary implementation

$$\Gamma$$

of the covering element $z = -1$, compatible with both the tensorial spin-cover action and the anchored rotational comparison of Section 6. In particular, on each comparison sector $\mathcal{H}_\rho$,

$$\Gamma|_{\mathcal{H}_\rho} = \widetilde{U}_\rho(-I). \quad (172)$$

Since $z^2 = 1$,

$$\Gamma^2 = I. \quad (173)$$

Because Γ is unitary,

$$\Gamma^\dagger = \Gamma. \quad (174)$$

Hence Γ is a self-adjoint unitary involution.

Its action on tensorial geometry follows directly from the covering map.

Proposition 7.1 (Tensorial blindness to the spin-cover center). *For every represented tensorial geometric field*

$$F \in \mathfrak{F}_{\text{geom}}^{\text{tens}},$$

the covering-kernel element acts trivially by conjugation:

$$\Gamma F \Gamma^{-1} = F. \quad (175)$$

In particular,

$$[\Gamma, A] = 0, \quad A \in \mathcal{A}_{\text{geom}}, \quad (176)$$

and therefore

$$\Gamma \in \mathcal{A}'_{\text{geom}}. \quad (177)$$

Proof. By assumption, the transformation law of every

$$F \in \mathfrak{F}_{\text{geom}}^{\text{tens}}$$

factors through

$$p : \text{Spin}^+(1, 3) \longrightarrow \text{SO}^+(1, 3).$$

Since

$$p(z) = I,$$

the element z induces the identity in every ordinary Lorentz vector and tensor representation. Hence

$$\Gamma F \Gamma^{-1} = F.$$

The same conclusion holds for the observable subalgebra $\mathcal{A}_{\text{geom}}$. □

This proposition supplies the link needed below: the central sign visible in the anchored projective comparison is invisible to the tensorial geometric algebra.

The spectral projections of Γ are

$$P_{\pm} = \frac{1}{2}(I \pm \Gamma). \quad (178)$$

They define orthogonal subspaces

$$\mathcal{H}_{\pm} := P_{\pm} \mathcal{H}, \quad (179)$$

on which

$$\Gamma|_{\mathcal{H}_{\pm}} = \pm I.$$

Since

$$[\Gamma, A] = 0$$

for every $A \in \mathcal{A}_{\text{geom}}$, the projections $P_{\pm}$ also commute with the geometric observable algebra.

Proposition 7.2 (Geometric reducing decomposition). *The subspaces*

$$\mathcal{H}_{+}, \quad \mathcal{H}_{-}$$

are reducing subspaces for $\mathcal{A}_{\text{geom}}$:

$$\mathcal{A}_{\text{geom}} \mathcal{H}_{\pm} \subseteq \mathcal{H}_{\pm}. \quad (180)$$

Equivalently,

$$[P_{\pm}, A] = 0 \quad \text{for all } A \in \mathcal{A}_{\text{geom}}. \quad (181)$$

Proof. Since $P_{\pm}$ are polynomials in Γ , Proposition 7.1 gives

$$[P_{\pm}, A] = 0.$$

If $\psi_{\pm} = P_{\pm} \psi$, then

$$A \psi_{\pm} = A P_{\pm} \psi = P_{\pm} A \psi \in \mathcal{H}_{\pm}.$$

□

Thus

$$\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_- \quad (182)$$

is a reducing decomposition relative to the represented tensorial geometric algebra whenever the central involution Γ exists.

It is useful to encode the covering character by

$$\varsigma_\rho(z) = (-1)^{\epsilon(\rho)}, \quad \epsilon(\rho) \in \mathbb{Z}_2. \quad (183)$$

Thus

$$\epsilon(\rho) = 0$$

denotes the tensorial covering character and

$$\epsilon(\rho) = 1$$

the nontrivial spin-cover character.

Under tensor products the central characters multiply, and therefore

$$\epsilon(\rho \otimes \sigma) = \epsilon(\rho) + \epsilon(\sigma) \pmod{2}. \quad (184)$$

Equivalently,

$$(+) \otimes (+) = (+), \quad (185)$$

$$(+) \otimes (-) = (-), \quad (186)$$

$$(-) \otimes (-) = (+). \quad (187)$$

These relations express multiplication of the two characters of the covering kernel; they are not a dynamical fusion law.

7.2 Comparison parity and relative sector separation

For the standard anchored orthogonal- π comparison protocol of Section 6, the comparison loop lifts to the nontrivial covering element

$$z = -1.$$

By Eq. (172), on an irreducible comparison sector,

$$\Xi_\rho = \widetilde{U}_\rho(z) = \Gamma_\rho. \quad (188)$$

For an irreducible spin- j rotational sector this becomes

$$\Xi_\rho = \Gamma_\rho = (-1)^{2j} I. \quad (189)$$

Thus the rephasing-invariant comparison diagnostic and the central grading seen by the geometric commutant are two realizations of the same covering-kernel character.

The statement is sectorwise. If a Hilbert space contains both covering characters, then

$$\Gamma = I_{\mathcal{H}_+} \oplus (-I_{\mathcal{H}_-}), \quad (190)$$

rather than a single scalar multiplier on the full reducible representation. The decomposition Eq. (182) keeps the two characters distinct.

To compare this grading with a geometry-generated representation, introduce one further assumption. Let

$$|\Omega\rangle$$

be a reference state that is even under the covering kernel,

$$\Gamma|\Omega\rangle = |\Omega\rangle. \quad (191)$$

This state is used only as an even tensorial reference state; no dynamical vacuum-selection statement is assumed.

Define the closed subspace generated by geometric observables,

$$\mathcal{H}_{\text{geom}} := \overline{\mathcal{A}_{\text{geom}}|\Omega\rangle}. \quad (192)$$

Since $\mathcal{A}_{\text{geom}}$ commutes with Γ ,

$$\mathcal{H}_{\text{geom}} \subseteq \mathcal{H}_+. \quad (193)$$

This gives the relative sector-separation statement.

Theorem 7.3 (Relative spin-cover sector separation). *Assume*

$$\Gamma|\Omega\rangle = |\Omega\rangle$$

and let $\mathcal{H}_{\text{geom}}$ be defined by Eq. (192). If a quantum comparison sector $\mathcal{H}_\rho$ carries the nontrivial covering character,

$$\Gamma|_{\mathcal{H}_\rho} = -I, \quad (194)$$

then

$$\mathcal{H}_\rho \cap \mathcal{H}_{\text{geom}} = \{0\}. \quad (195)$$

Moreover, no operator in $\mathcal{A}_{\text{geom}}$ maps a nonzero vector in $\mathcal{H}_+$ into $\mathcal{H}_-$.

Proof. Equation (193) gives

$$\mathcal{H}_{\text{geom}} \subseteq \mathcal{H}_+,$$

whereas Eq. (194) gives

$$\mathcal{H}_\rho \subseteq \mathcal{H}_-.$$

Since Γ is self-adjoint, its $+1$ and -1 eigenspaces are orthogonal. Their intersection is therefore $\{0\}$, which proves Eq. (195).

For the second statement, let

$$\psi_+ \in \mathcal{H}_+, \quad A \in \mathcal{A}_{\text{geom}}.$$

Proposition 7.2 implies

$$A\psi_+ \in \mathcal{H}_+.$$

Hence $A\psi_+$ cannot be a nonzero vector in $\mathcal{H}_-$. □

The theorem establishes a reducing decomposition relative to the specified tensorial geometric observable algebra. It is weaker than a superselection theorem in the algebraic quantum-field-theory sense. A Doplicher–Haag–Roberts interpretation would require a local observable net, localized and transportable representations, appropriate causality and duality properties, and the hypotheses of the corresponding reconstruction theorems [8, 9, 10].

A direct consequence is that if

$$\psi_+ \in \mathcal{H}_+$$

and an operator O produces a state with a nonzero component in $\mathcal{H}_-$, then

$$[\Gamma, O] \neq 0 \tag{196}$$

on the relevant domain. Such an operator cannot belong to $\mathcal{A}_{\text{geom}}$. Any mechanism connecting the two covering parities must therefore enlarge the purely tensorial geometric observable algebra.

This statement does not determine the required enlargement. It does not derive a matter creation operator, charged field algebra, defect algebra, or internal gauge algebra. Likewise,

$$(-) \otimes (-) = (+)$$

expresses only the multiplication of covering characters; it does not imply dynamical pair production.

The covering parity should also not be identified with fermion parity. The relation

$$\Xi_\rho = \Gamma_\rho = -I$$

identifies the nontrivial spin-cover character. The relativistic spin–statistics connection requires additional assumptions including locality, covariance, positivity, and a suitable spectrum condition [13]. These structures are absent from the fiberwise quantum kinematics considered here.

Nor does the result select a particular half-integer representation. The condition

$$\Gamma = -I$$

is compatible with

$$j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$$

Selecting $j = \frac{1}{2}$ requires an independent minimality or dynamical principle. Constructing Weyl or Dirac covariant field representations likewise requires an additional relativistic field-theoretic realization.

The local soldering sector and the global comparison sector may now be displayed side by side:

$$\begin{array}{c} \text{operational response} \\ \Theta_{\Pi, \xi} \end{array} \longrightarrow \mathcal{E}_x \longrightarrow T^* \mathcal{E}_x \longrightarrow \text{local Weyl quantum kinematics}, \tag{197}$$

while independently,

$$\begin{array}{c} \text{anchored observer comparison} \\ SO(3) \end{array} \longrightarrow \Xi_\rho = \Gamma_\rho \longrightarrow \mathbb{Z}_2 \text{ comparison parity}. \tag{198}$$

The first chain originates in the cotangent completion of the operational soldering data. The second originates in the global topology of the finite rotational comparison problem. Their intersection

does not constitute a derivation of matter from geometry. It identifies the precise algebraic condition under which nontrivial projective comparison information is invisible to a tensorial geometric observable algebra.

The result is conditional in two independent respects. The operational soldering construction does not force the existence of a nontrivial $\Gamma = -I$ sector, and the tensorial geometric algebra does not supply operators that connect the two covering parities. Both require additional comparison or dynamical structure.

8 Discussion and Conclusions

The present paper studies the quantum kinematics associated with variations of an operationally reconstructed solder map. The classical input is supplied by Refs. [15, 16]: the first selects the relativistic massive-generator geometry, while the second reconstructs a nondegenerate solder map through

$$\tilde{\Theta}_x(\Pi, \xi) = \Pi(e_x \xi), \quad e_x : T_x M \xrightarrow{\sim} V_x, \quad (199)$$

with

$$g = e^* \eta.$$

The present construction begins when this reconstructed solder map is allowed to vary.

8.1 Main results and logical status

At each event the ambient soldering configuration space is

$$\mathcal{E}_x = \text{Hom}(T_x M, V_x). \quad (200)$$

For

$$\Pi \in V_x^*, \quad \xi \in T_x M,$$

the incidence expression becomes the linear configuration observable

$$\Theta_{\Pi, \xi}(e) = \Pi(e\xi). \quad (201)$$

Since

$$\mathcal{E}_x^* \simeq V_x^* \otimes T_x M,$$

these functions span the full linear dual and separate points of $\mathcal{E}_x$. The underlying tensor-product identity is standard; the relevant structural fact is that this complete configuration-observable family is inherited from the same operational response that reconstructed the classical solder map.

The first quantum result is a boundary theorem. If the original generator and displacement composition laws are implemented exactly and independently at the unitary level, all local response unitaries commute. The exact-response algebra is therefore Abelian. Its spectral parameters can be represented by linear solder maps, but the algebra contains no operator that coherently changes the soldering configuration.

Noncommutative kinematics appears only after an additional input is introduced. The coherent soldering-deformation principle assumes that

$$e \mapsto e + A, \quad A \in \mathcal{E}_x, \quad (202)$$

may be implemented coherently. Adopting the standard cotangent completion

$$\mathcal{P}_x = T^* \mathcal{E}_x$$

then introduces

$$\mathbf{P} \in \mathcal{E}_x^*$$

as the momentum conjugate to e . If $K(A)$ generates the translation by A ,

$$[\widehat{K}(A), \widehat{\Theta}_{\Pi, \xi}] = i\hbar \Pi(A\xi)I. \quad (203)$$

The response observables themselves remain mutually commuting. Noncommutativity therefore arises from the coherent configuration-changing operators rather than from a modification of the original incidence law. This distinction is central:

$$\text{exact response} \implies \text{Abelian algebra,}$$

whereas

$$\text{response} + \text{coherent soldering deformations} \implies \text{Weyl quantum kinematics.}$$

The cotangent and Weyl constructions themselves are standard. Their role here is fixed by the operational provenance of the configuration observables and by the pairing

$$(\Pi, \xi; A) \mapsto \Pi(A\xi).$$

Internal Lorentz reduction confirms the geometric interpretation of the pre-reduction variables. On the nondegenerate sector,

$$\mathcal{E}_x^\times / SO^+(1, 3) \simeq \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M), \quad (204)$$

and at zero internal Lorentz momentum,

$$J^{-1}(0) / SO^+(1, 3) \simeq T^* \text{Met}_{1,3}^{\text{or}, \uparrow}(T_x M). \quad (205)$$

Thus the operationally selected soldering variables reduce to the ordinary local metric configuration variables rather than describing an unrelated sixteen-component system.

The finite observer-comparison construction is logically separate. Once an anchored rotational comparison sector is supplied, the $SO(3)$ action on quantum rays may be projective. For the standard orthogonal- π protocol,

$$\Xi_\rho = U_\rho(r_1)U_\rho(r_2)U_\rho(r_1)^{-1}U_\rho(r_2)^{-1} \quad (206)$$

is rephasing invariant and, in an irreducible lifted sector,

$$\Xi_\rho = \varsigma_\rho(-I)I. \quad (207)$$

The two possible signs distinguish the two rotational spin-cover characters.

This projectivity has a different origin from the Weyl phase:

$$\begin{aligned} \text{local canonical projectivity} &\longleftrightarrow \text{symplectic pairing on } T^* \mathcal{E}_x, \\ \text{global rotational projectivity} &\longleftrightarrow \pi_1(SO(3)) \simeq \mathbb{Z}_2. \end{aligned} \quad (208)$$

Canonical noncommutativity therefore does not imply nontrivial spin-cover projectivity.

The covering element

$$z = -1$$

projects to the identity in the ordinary Lorentz group and acts trivially on vector and tensor representations. If its implementation is denoted by

$$\Gamma,$$

then

$$\Gamma \in \mathcal{A}'_{\text{geom}}. \quad (209)$$

Hence the $\Gamma = \pm 1$ eigenspaces are reducing subspaces for the tensorial geometric observable algebra. Under the even-reference assumption of Section 7, a comparison sector with

$$\Gamma_\rho = \Xi_\rho = -I$$

is disjoint from the geometry-generated even reference sector. This is a relative spin-cover sector-separation statement, not a DHR superselection theorem.

The principal local chain may therefore be summarized as

$$\begin{array}{c} \text{operational response} \\ \Theta_{\Pi, \xi} \end{array} \longrightarrow \mathcal{E}_x \longrightarrow T^*\mathcal{E}_x \longrightarrow \text{Weyl quantum kinematics}, \quad (210)$$

where the cotangent step requires the separately stated coherent soldering-deformation principle.

The finite comparison branch is instead

$$\begin{array}{c} \text{anchored rotational comparison} \\ SO(3) \end{array} \longrightarrow \Xi_\rho = \varsigma_\rho(-I)I \longrightarrow \mathbb{Z}_2 \text{ covering parity}. \quad (211)$$

The second chain is conditional and secondary to the local soldering quantum kinematics.

8.2 Scope and open problems

The common variable at the classical–quantum interface is the solder map e . Its two uses may be displayed schematically as

$$\widetilde{\Theta} \longrightarrow e \begin{cases} \longrightarrow g = e^*\eta \longrightarrow \text{restricted classical gravitational completion,} \\ \longrightarrow \mathcal{E}_x \longrightarrow T^*\mathcal{E}_x \longrightarrow \text{local quantum soldering kinematics.} \end{cases} \quad (212)$$

This identifies operationally reconstructed soldering as a natural kinematical interface between Lorentzian geometry and canonical quantum structure. It does not yet provide a quantum dynamics of gravity.

The first unresolved problem is the passage from the fiberwise space $\mathcal{E}_x$ to a field configuration space of soldering sections. Such a theory requires an infinite-dimensional symplectic structure, regularization, operator domains, boundary conditions, and a treatment of internal Lorentz and spacetime diffeomorphism constraints. The local finite-dimensional Stone–von Neumann result does not extend unchanged to this setting.

The second unresolved problem is dynamics. No Hamiltonian or action on soldering configuration space is selected here. In particular, the cotangent momentum

$$\mathbf{P} \in T_e^*\mathcal{E}_x$$

has not been identified with a specific gravitational connection, extrinsic-curvature variable, or first-order gravitational momentum. Relating the present local cotangent space to a constrained gravitational phase space requires additional dynamical input.

The third problem concerns finite comparison. The topology of $SO(3)$ permits

$$\Xi_\rho = -I,$$

but does not require a nontrivial comparison sector to occur. A local frame rotation remains gauge unless the relevant comparison histories are operationally anchored. A fuller theory must therefore specify both the physical comparison protocol and the dynamics, if any, that selects its projective class.

A fourth problem is the passage from spin-cover comparison parity to a localized relativistic matter theory. The present result establishes only

$$\Gamma \in \mathcal{A}'_{\text{geom}}$$

and the corresponding reducing decomposition. It does not derive a spinor field, an internal gauge group, a matter spectrum, or particle statistics. Localization, transportability, causal composition, positive-energy structure, and an appropriate field or observable algebra are required before a DHR-type or spin–statistics interpretation becomes available [8, 9, 10, 13].

Finally, the classical and quantum branches remain dynamically unconnected. A field theory would have to determine how soldering fluctuations affect the reconstructed metric, how gravitational constraints act on the enlarged quantum algebra, and whether nontrivial comparison sectors couple to geometric degrees of freedom. None of these structures follows from the fiberwise kinematics proved here.

The principal conclusion is therefore limited but precise. The operational generator–displacement response does more than reconstruct one classical solder map: when the map is allowed to vary, the same incidences supply a complete linear family of soldering configuration observables. Exact implementation of their classical composition laws remains Abelian. Once coherent soldering deformations are supplied as an additional principle, the standard cotangent completion gives the canonical noncommutative relation

$$[\widehat{K}(A), \widehat{\Theta}_{\Pi, \xi}] = i\hbar \Pi(A\xi)I.$$

This operationally selected local Weyl kinematics is the central result of the paper. Global spin-cover comparison provides a distinct, conditional extension rather than its foundation.

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